

AFL Data Foundations — Data Dictionary & Target Definitions

Web3Geek Internship · Week 3, Day 1 · Fiza Aslam

1. Source Tables

Table	Grain	Key Columns	Rows
player_info	one row per player	id	2,848
round_stats	one row per player per match	player_id, id (game), match_date	274,079
season_stats	one row per player per season	player_id, year, is_finals	25,491
team_matches	one row per team per match (long format — 2 rows/match)	team_name, match_date, opponent	15,808

Join logic: round_stats.player_id → player_info.id. No shared numeric match_id exists between round_stats and team_matches — matches are linked via a synthetic key of (match_date, team, opponent).

2. Key Data Quality Findings

- team_matches team names were whitespace/tab-corrupted (40 raw values → 20 real teams after cleaning).
- round_stats.score is 100% NULL — fantasy_points (official, dataset-provided) used as the composite metric instead.
- round_stats & season_stats each had 10 exact duplicate rows; player_info had 5 — all dropped.
- Advanced stats (hit_outs, bounces, contested_marks, brownlow_votes) are structurally missing pre-availability years (e.g. bounces/contested_marks only from 2003, brownlow_votes only from 2006) — not random missingness.
- Coverage: 1983–2025, 20 teams, 2,843 unique players after de-duplication.

3. Prediction Target Definitions

Target	Definition	Formula	Level
target_result	Match outcome (primary, classification)	W/L/D on home-team row → Home Win / Away Win / Draw	match
margin_signed	Signed score differential (secondary, regression)	+margin if home won, –margin if away won	match
is_top_disposals_game	Leading disposal-getter in a match	rank(disposals) within game id, top = 1	player-game
is_top_goals_game	Leading goal-kicker in a match	rank(goals) within game id, top = 1	player-game
fantasy_points	Composite performance score (official)	dataset-provided AFL Fantasy points column	player-game

Framing rationale: classification is primary since predicting the winning side is the core deliverable; draws are rare (<1% of matches), so heavy class imbalance is expected on that class. margin_signed supports confidence/spread-style predictions.

4. Engineered Features (leakage-safe — all use .shift(1) before .rolling()/expanding())

Feature	Description	Window	Source
last5_avg_score / last3_win_rate / last5_win_rate	Team's rolling average score / win rate over previous games	rolling 3–5, shifted	team_score, win
win_streak	Current consecutive win(+)/loss(–) streak entering the match	expanding, shifted	win
days_rest	Days since team's previous match	diff	match_date
h2h_win_rate	Historical win rate vs this specific opponent	expanding, shifted	win, matchup
venue_win_rate	Team's historical win rate at this venue	expanding, shifted	win, venue
ladder_rank_before_match	League ladder position entering the match	cumulative, shifted	win, year
last5_avg_disposals / _goals / _fantasy	Player's rolling average output over last 5 games	rolling 5, shifted	round_stats player cols

Saved as: features_team_v1.csv (15,808 rows), features_player_v1.csv (274,079 rows).

5. Train/Hold-Out Split & Accuracy Ceiling

Split: strict time-based split via a reusable get_time_split() function — holds out the most recent season (or last N rounds) rather than a random split, since rolling/H2H/ladder features are computed from each team's match history; a random split would let the model see "future" games in training that inform an "earlier" test match, inflating performance in a way that would never hold for genuinely upcoming matches.

Realistic ceiling: published AFL win-prediction models typically reach ~65–75% classification accuracy, against a home-favorite baseline already at ~59% (this dataset's observed home win rate). Match outcomes carry irreducible noise (injuries, umpiring, on-the-day form) that pre-match features can't fully capture. A holdout accuracy near 95%+ would be a strong red flag for leakage rather than genuine skill.